DX-SMART

DX-LR30-900M22S

Module technical manual

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Updated records

Version	Date	Instructions	Author
V1.0	2024/11/18	Initial version	SML
V2.0	2024/12/17	Optimizing RF parameters	SML

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Contents

Form Index - 16.

Image index

1. Module Introduction

1.1. Overview

DX-LR30-900M22S is a low-power ultra-long distance LoRa pure RF hardware module, 850-930MHz frequency band use, is built by SHEN ZHEN DX-SMART TECHNOLOGY CO.,LTD,. for intelligent wireless data transmission, using SX1262 RF chip, integrated power amplifier, anti-interference performance and communication distance are further improved, Has the advantages of low power consumption, high performance and long distance.

Because the module is pure RF hardware, it needs to use MCU driver or use special SPI debugging tool.

1.2. Features

Performance:

- Output power: +22dBm
- Maximum receiving sensitivity: -125dBm (BW=125, SF=7)
- Open visible distance up to 8km (for reference only, the actual distance is subject to the actual measurement)

Module parameters:

- Operating voltage: 1.8V-3.7V (typical value: 3.3V)
- Support working frequency range: 850-930MHz
- External antenna
- Operating temperature: -40~+85 ° C

Note: The current module uses a non-temperature compensated crystal oscillator. When the bandwidth is below 125K, it may cause excessive frequency offset. If the bandwidth is below 125K, please contact our business personnel for customization.

1.3. Basic parameters

Table 1: Table of basic parameters1

<table>
<tr><th>Parameter names</th><th>Description</th><th colspan="2">Notes</th></tr>
<tr><td rowspan="4">Reference distance</td><td rowspan="4">8km</td><td rowspan="4">Clear open environment, antenna height 3 m, air rate 244bit/s</td><td>SF:12</td></tr>
<tr><td>CR:4/6</td></tr>
<tr><td>BW:125</td></tr>
<tr><td>frequency band:868/915</td></tr>
<tr><td>FIFO</td><td>256Byte</td><td colspan="2">Maximum length for a single send</td></tr>
<tr><td>Crystal frequency</td><td>32MHz</td><td colspan="2">-</td></tr>
<tr><td>Modulation</td><td>LoRa</td><td colspan="2">LoRa modulation is recommended</td></tr>
<tr><td>Encapsulation method</td><td>Patch type</td><td colspan="2">-</td></tr>
<tr><td>Interface method</td><td>Stamp hole</td><td colspan="2">-</td></tr>
<tr><td>Communication interface</td><td>SPI</td><td colspan="2">0~10Mbps</td></tr>
<tr><td>Dimensions</td><td>20.0(L) x 14.0 (W) x 2.3(H) mm</td><td colspan="2">Shield hood included</td></tr>
<tr><td>Rf interface</td><td>Stamp hole /IPEX</td><td colspan="2">-</td></tr>
</table>

1.4. Working parameters

Table 2: Work Parameter Sheet2

Parameter names	Details			Notes
	Minimum	Typical value	Maximum value	
Operating voltage (V)	1.8	3.3	3.7	≥3.3V ensures output power

Communication level (V)		-	3.3	-	Use a 5V TTL with a risk of burning
Operating temperature (C)		-40	-	85	Industrial grade design
Operating frequency band (MHz)		850	868/915	930	Supports ISM
Power consumption	Transmit current (mA)	-	100	-	Instantaneous power consumption
	Received current (mA)	-	10	-	-
	Standby current (nA)	-	180	-	Software turn-off
transmit power (dBm)		0	-	22	-
Receiver sensitivity (dBm)		-	-125	-	BW=125, SF=7
Air rate	LoRa(bps)	-	-	-	User programmed control

2. Application interface

2.1. Module pin definition

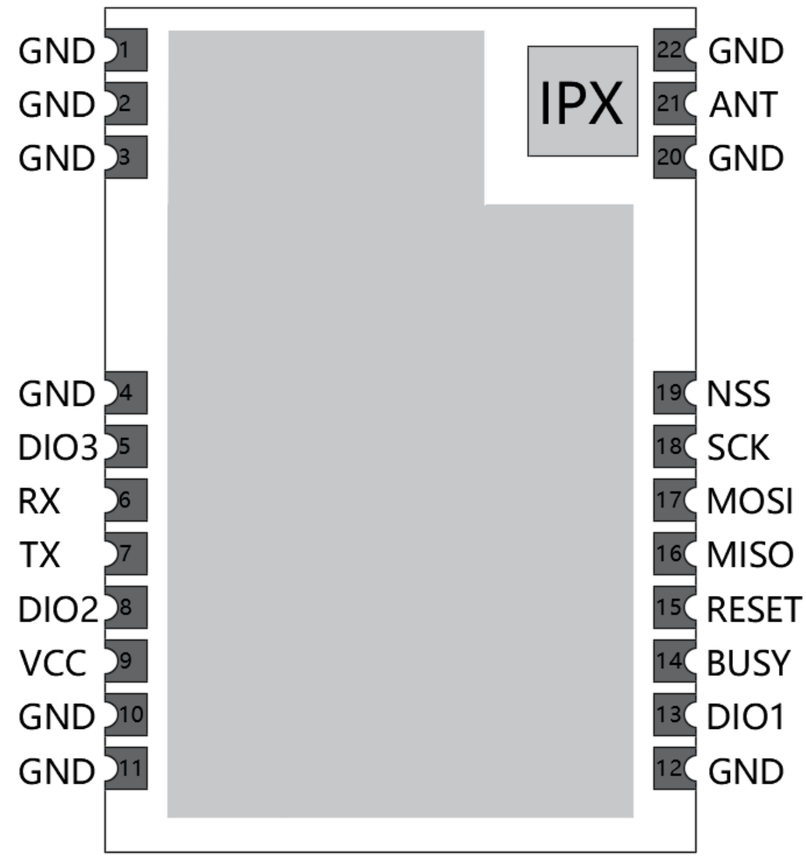


Figure 1: Module pin definition1

2.2. Illustration of the pin definition

Table 3: Pin Definition Instructions Table 3

Pin serial number	Pin name	Pin function	Instructions

1,2,3,4,10,11,12,20,22	GND	Power ground	-
5	DIO3	Multi-function digital input/output - External TCXO power supply voltage	Input/output
6	RX	Antenna toggle switch	Used to switch receiving and transmitting
7	TX	Antenna toggle switch	Used to switch receiving and transmitting
8	DIO2	Multipurpose digital input/output/RF band switching control	Input/Output
9	VCC	Power input pin	3.3V(typical value)
13	DIO1	Multipurpose digital input/output	Input/output
14	BUSY	Used for status indication	-
15	RESET	Foot reset	-
16	MISO	SPI data output pin	-
17	MOSI	SPI data input pin	-
18	SCK	SPI clock input pin	-
19	NSS	Module pieces select pins for starting an SPI communication	-
21	ANT	Rf interface	-

2.3. Reference Basic circuit

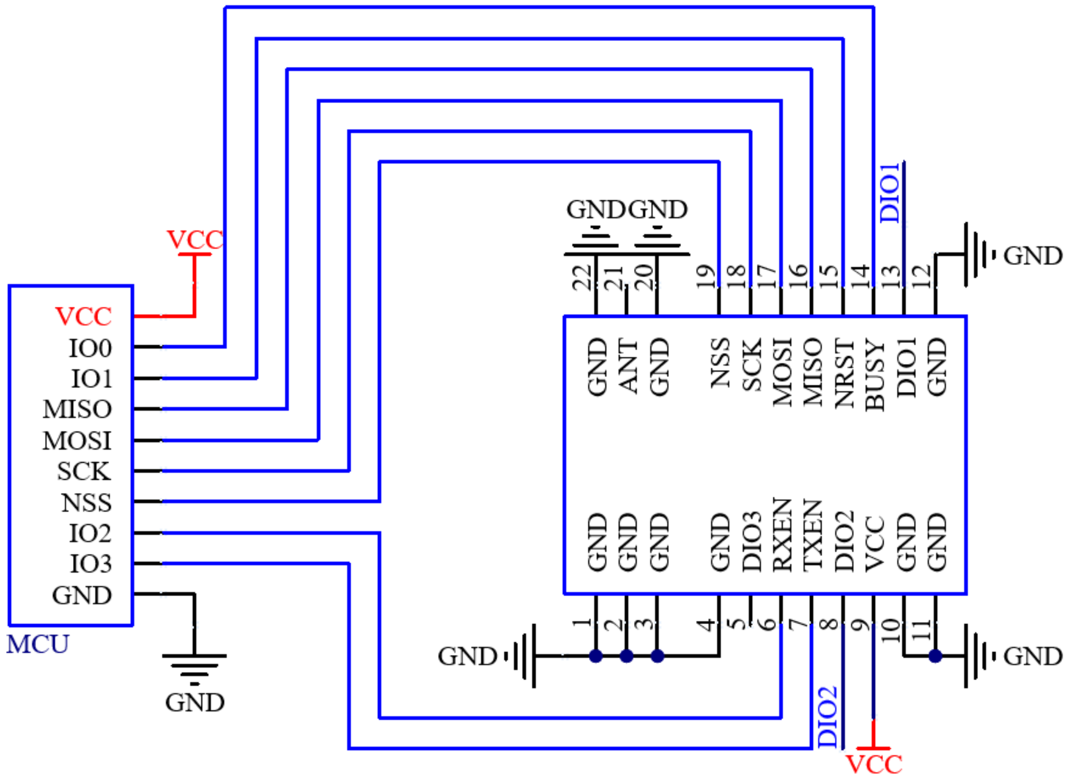


Figure 2: Basic circuit2

3. Electrical characteristics, RF characteristics, and reliability

3.1. Maximum rating

Pressure in excess of the absolute maximum rating may cause permanent damage to the equipment. These are stress ratings only and therefore do not imply functional operation of the equipment under these or any other conditions beyond those indicated in the operating section of the instruction manual. Prolonged exposure to absolute maximum rating conditions may affect the reliability of the equipment.

Table 4: Table of absolute maximum ratings4

Parameters	Minimum	Maximum	Units
VBAT	-0.5	3.7	V
Storage temperature range	-55	+125	°C

Table 5: Recommended conditions of use5

Parameters	Minimum	Typical value	Maximum value	Units
VBAT	1.8	3.3	3.7	V
Operating temperature range (TA)	-40	-	+85	°C

3.2. Static protection

In the application of modules, due to the static electricity generated by human body static electricity and charged friction between microelectronics, it may cause some damage to the module through various ways, so ESD protection should be paid attention to. ESD protection measures should be taken in the process of research and development, production, assembly and testing, especially in product design. For example, at the interface of the circuit design and the point susceptible to electrostatic discharge damage or influence, anti-static protection should be increased, and anti-static gloves should be worn in production.

Table 6: Table of ESD tolerant voltage of module pins6

Symbols	Description	Max	Units
ESD_HBM	American National Institute for Standardization/Electronic Systems Design Association/Associated Electronic Equipment Manufacturing Chamber of Commerce Standard JS-001-2014 (mannequins) Category 2	+2	kV
ESD_CDM	ESD charging device model, JEDEC standard JESD22-C101D, three levels	+1	kV

4. Basic operations

4.1. Hardware design

- It is recommended to use DC regulated power supply to power the module, and the power ripple coefficient should be as small as possible, and the module should be reliably grounded.
- Please pay attention to the correct connection of the positive and negative poles of the power supply, such as reverse connection may lead to permanent damage to the module;
- Please check the power supply, ensure the recommended between the power supply voltage, such as more than the maximum permanent damage will cause the module;
- Please check the stability of the power supply, the voltage can not fluctuate greatly and frequently;
- When designing the power supply circuit for the module, it is often recommended to reserve more than 30% of the margin, which is conducive to the long-term stable work of the whole machine;
- Module should be running away from the power source, transformer, high frequency electromagnetic interference such as larger part;
- High frequency digital go line, high frequency analog line, power line must avoid modules below, if you really need to pass beneath the module to suppose module welded on the Top Layer, the module contact part of the Top Layer of floor copper (copper and entire store good grounding), must be close to the module part Numbers and line on Bottom Layer;
- Assuming that the module is solder or placed in the Top Layer, in the Bottom Layer or other layers of random wiring is also wrong, will affect the spurious and receiving sensitivity of the module in different degrees;
- Suppose around module have larger electromagnetic interference devices would greatly influence the performance of the module, according to the strength of the interference advice from appropriate modules, if conditions allow can do appropriate isolation and blocking.

- It is assumed that there is a large electromagnetic interference around the module (high-frequency digital, high-frequency analog, power supply line) will also greatly affect the performance of the module, according to the strength of the interference, it is recommended to stay away from the module, if the situation permits, appropriate isolation and shielding can be done;
- Telecommunication lines if use 5 v level, must be 1 k to 5.1 k series resistor (not recommended, there is still a risk of damage);
- Try to stay away from TTL protocols that have a physical layer of 2.4GHz, such as USB3.0;
- Antenna installation structure has a great impact on the performance of the module, be sure to ensure that the antenna is exposed, preferably vertical. When the module is installed inside the shell, high-quality antenna extension cable can be used to extend the antenna to the outside of the shell;
- Antenna should not be installed within the metal casing and will lead to the transmission distance greatly weakened;
- It is recommended to add 200R protection resistor to the RXD/TXD of the external MCU.

4.2. Software writing

- DIO1, DIO2 and DIO3 is generally general IO mouth, can be configured to a variety of functions; Among them, DIO2 can be connected with TXEN, not with MCU IO port connection, used to control RF switch emission, see SX1262 manual, if not used can be suspended;
- SX1262 / SX1268 LLCC68 difference:

  SX1262/SX1268 supports spreading factors SF5, SF6, SF7, SF8, SF9, SF10, SF11, SF12;

  LLCC68 support spreading factor SF5, SF6, SF7, SF8, SF9, SF10, SF11.

5. Mechanical dimensions and layout suggestions

This section describes the mechanical dimensions of the module, all dimensions are in millimeters; All dimensions not marked with tolerances with tolerances of ±0.3 mm.

5.1. Module machinery '

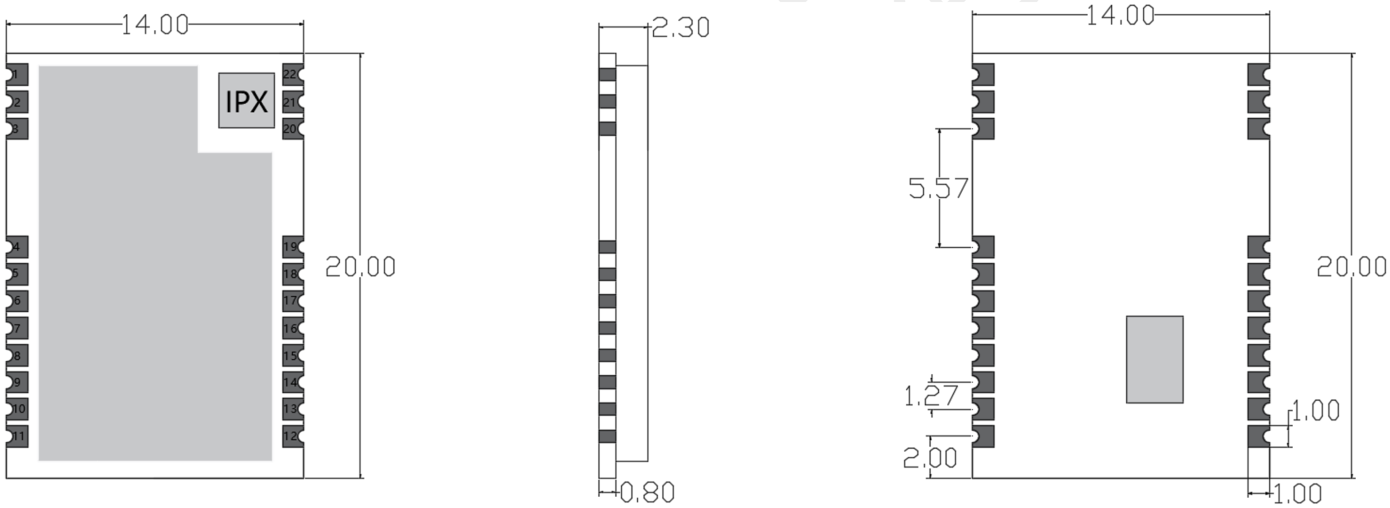


Figure 3: Module size chart for top view, side view and bottom view3

5.2. Recommended packaging

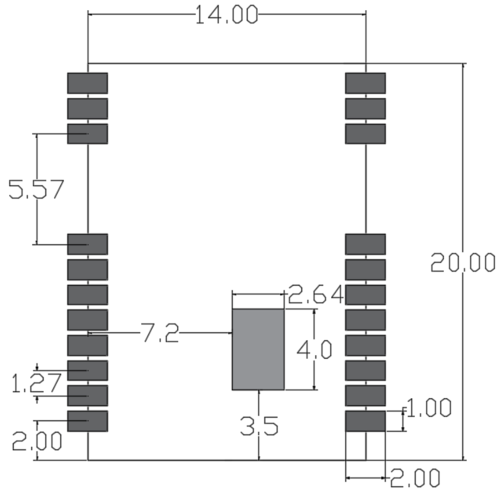


Figure 4: Bottom view of recommended package size4

5.3. Module top view/bottom view

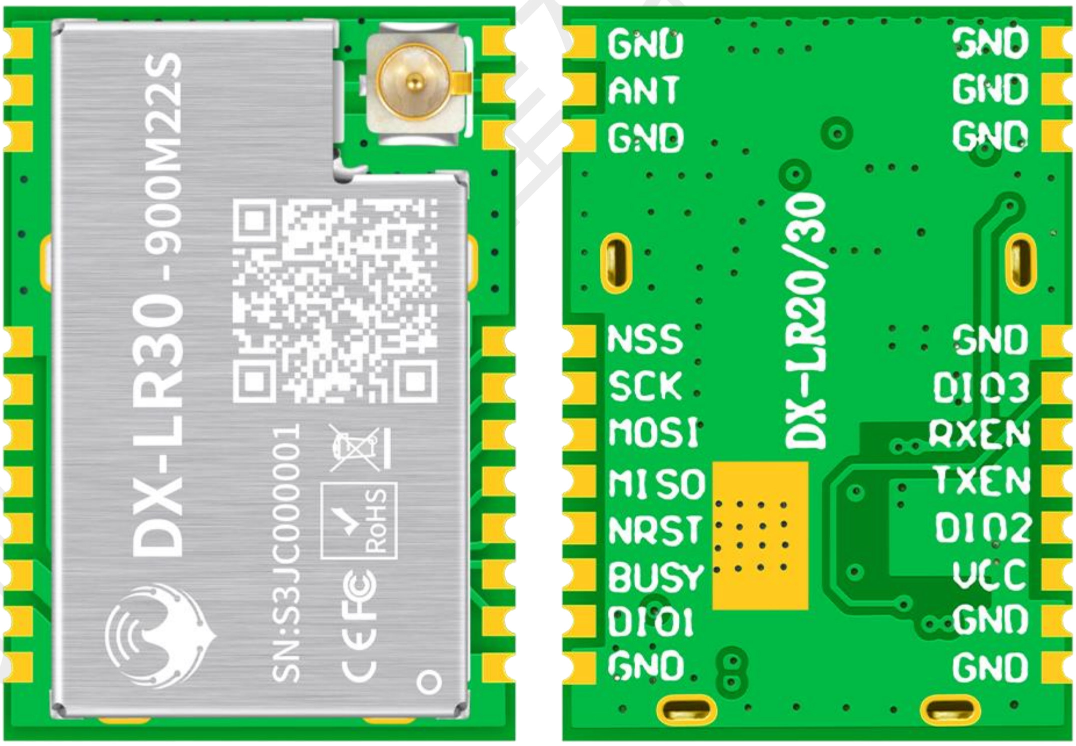


Figure 5: Module top view and bottom view5

note

Above is for reference only, actual product appearance and label information, please refer to the module object.

5.4. Hardware design layout suggestions

DX - LR30-900 m22s module working at 850 m - 930m frequency, using the external antenna, the antenna standing wave ratio pattern (VSWR) and efficiency depends on the patch position, should try to avoid the influence of various factors on the wireless sending and receiving signals, pay attention to the following:

1, surrounding the LR30-900M22S product shell to avoid the use of metal, when using part of the metal shell, should try to let the module antenna part away from the metal part. Metal wire or metal screws inside the product, should as far as possible away from the antenna module.

2. The module antenna part should be placed on the edge of the carrier board PCB or directly exposed to the carrier board, try not to be placed in the middle of the board.

3, Suggestions on base board module mounting location use insulating materials in isolation, for example, in the position to put a whole piece of silk screen (TopOverLay).

6. Q&A

6.1. Transmission distance is not ideal

- When there is a linear communication barrier, the communication distance will attenuate accordingly;
- Temperature, humidity, the same frequency interference, will lead to increased communication packet loss rate;

- The ground absorbs and reflects radio waves, and the test effect is poor near the ground.
- Seawater has a strong ability to absorb radio waves, so the seaside test effect is poor.
- There are metal objects near the antenna, or placed in a metal shell, the signal attenuation will be very serious;
- The power register is set incorrectly, the air rate is set too high (the higher the air rate, the closer the distance);
- Power supply voltage is lower than the recommended value at room temperature, the lower the voltage transmission power is smaller;
- Using antenna with poor module matching degree or antenna itself quality problem.

6.2. Module is easy to damage

- Please check the power supply to ensure that between the recommended supply voltage, if exceed the maximum value will cause permanent damage to the module;
- Please check the stability of the power supply, the voltage can not fluctuate greatly and frequently;
- Please ensure the anti-static operation during installation and use, and the electrostatic sensitivity of high-frequency devices;
- Please ensure that the installation process of humidity is unfavorable and exorbitant, some components for humidity sensitive devices;
- If there is no special demand, it is not recommended to use at too high or too low temperature.

6.3. Bit error rate is too high

- Near the same frequency interference, away from the interference sources or change frequency, channels to avoid interference;
- On the SPI clock waveform is not standard, to check whether there is interference, SPI line SPI bus go line shoulds not be too long;
- Power supply is not ideal also may cause garbled, be sure to guarantee the reliability of the power supply;
- Extension cord, the poor quality of feeder, or too long, can cause error rate is on the high side.

7. Storage, production and packaging

7.1. Storage conditions

Modules are shipped in vacuum-sealed bags. Humidity sensitive level for module 3 (MSL) 3, stores it needs to follow the following conditions:

1. The recommended storage conditions: temperature of 23±5°C, and relative humidity of 35 ~ 60%.

2. Under the recommended storage conditions, module can be in vacuum sealed bags for 12 months.

3. Under the workshop conditions of temperature 23±5°C and relative humidity below 60%, the workshop life of the module after unpacking is 168 hours. Under these conditions, the module can be directly used for reflux production or other high temperature operation. Otherwise, it is necessary to store the module in an environment with relative humidity less than 10% (for example, a moistureproof cabinet) to keep the module dry.

4. If the module is under the following conditions, it is necessary to pre-bake the module to prevent the PCB blistering, cracking and delamination after the module is hygro-absorbed and then welded at high temperature:

- Storage temperature and humidity do not meet the recommended storage conditions
- The module fails to complete production or storage according to Clause 3 above after unpacking
- Vacuum packaging leakage, materials in bulk
- Before module repair

7.2. Module baking treatment

- It needs to be baked at 120±5°C for 8 hours at high temperature
- The second baked module must be welded within 24 hours after baking, otherwise it still needs to be stored in the drying oven

Notes

1. In order to prevent and reduce the occurrence of bad welding such as foaming and delimitation caused by moisture, the module should be strictly controlled. It is not recommended to be exposed to the air for a long time after opening the vacuum package.
2. Before baking, it is necessary to remove the module from the package and place the bare module on the high temperature resistant appliance to avoid high temperature damage to the plastic tray or reel; The second baking module must be completed within 24 hours after baking welding, otherwise it needs to be stored in the drying oven. Please pay attention to ESD protection when unpacking and placing the module, for example, wear anti-static gloves.

7.3. Reflow soldering

Use a printing scraper to print solder paste on the screen plate, so that the solder paste is leaked to the PCB through the opening of the screen plate, and the printing scraper strength needs to be adjusted appropriately. In order to ensure the quality of the module paste, the thickness of the steel mesh corresponding to the module pad part is recommended to be 0.1~0.15mm.

The recommended reflow soldering temperature is 235~250 °C, and the maximum temperature should not exceed 250 °C. To avoid damage to the module due to repeated heat exposure, it is highly recommended that customers do not attach the module until the first side of the PCB has been reflow soldered. The recommended furnace temperature curve (lead-free SMT reflow soldering) and related parameters are shown in the following chart:

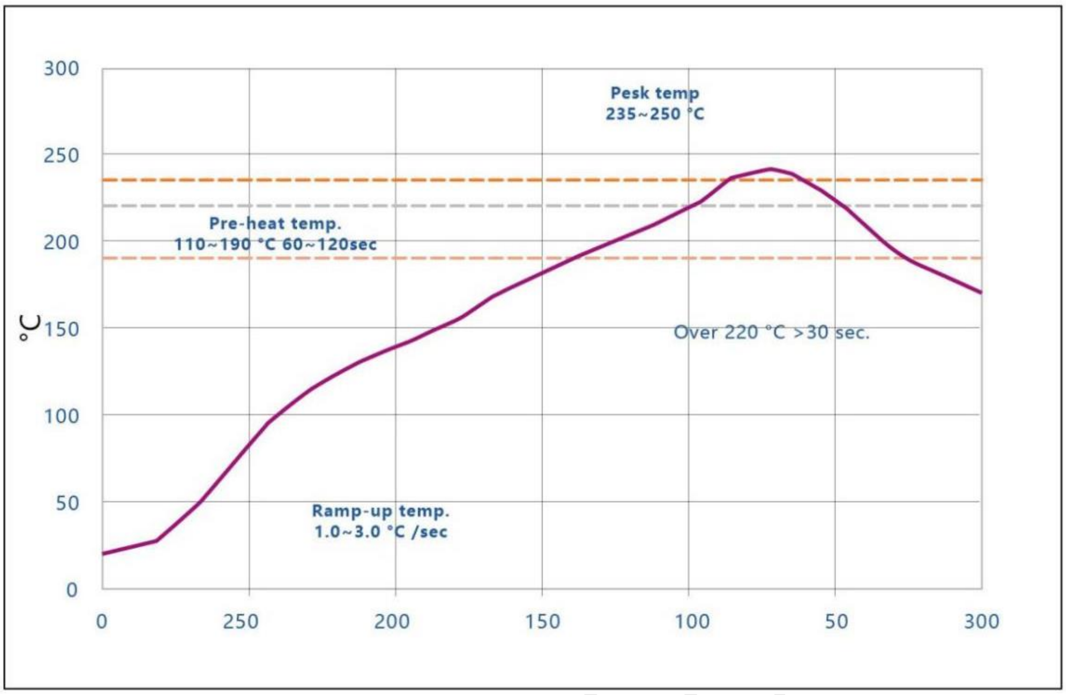


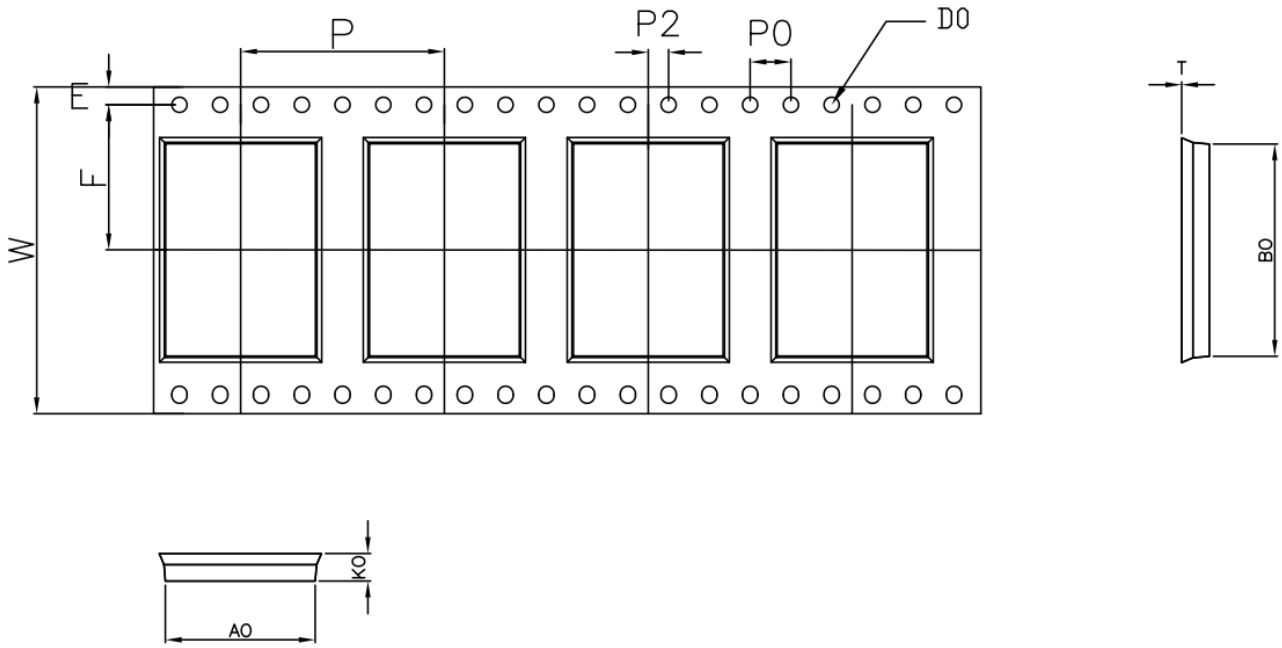
Figure 6: Recommended reflow soldering temperature curve 6

Table 7: Recommended reflow soldering temperature7

Statistical Name	Lower limit	Upper limit	Units
Slope 1 (target =2.0) is between 30.0 and 70.0	1	3	Degrees/SEC
Slope 2 (target =2.0) is between 70.0 and 150.0	1	3	Degrees/SEC
Slope 3 (target =-2.8) is between 220.0 and 150.0	-5	-0.5	Degrees/SEC
Constant temperature time 110-190 ° C	60	120	seconds
@220C reflux time	30	65	seconds
Peak temperature	235	250	Celsius
@235C total time	10	30	seconds

7.4. Packing specifications

The DX-LR30-900M22S module is packaged in roll tape and enclosed in a vacuum sealed bag with desiccants and humidity cards. Each carrier is 20.4 meters long, contains 1200 modules, and the reel is 330 mm in diameter. The specifications are as follows:



W	32.00±0.30	P	20.00±0.10	A0	14.30±0.10	B0	20.40±0.10
S0	————	P0	4.00±0.10	A1		B1	
E	1.75±0.10	P2	2.00±0.10	A2		B2	
F	14.2±0.10	D0	$\phi 1.50^{+0.10}_{-0}$	K0	2.70±0.10		
T	0.30±0.05	D1		K1			

Figure 7: Strap size (unit: mm)7

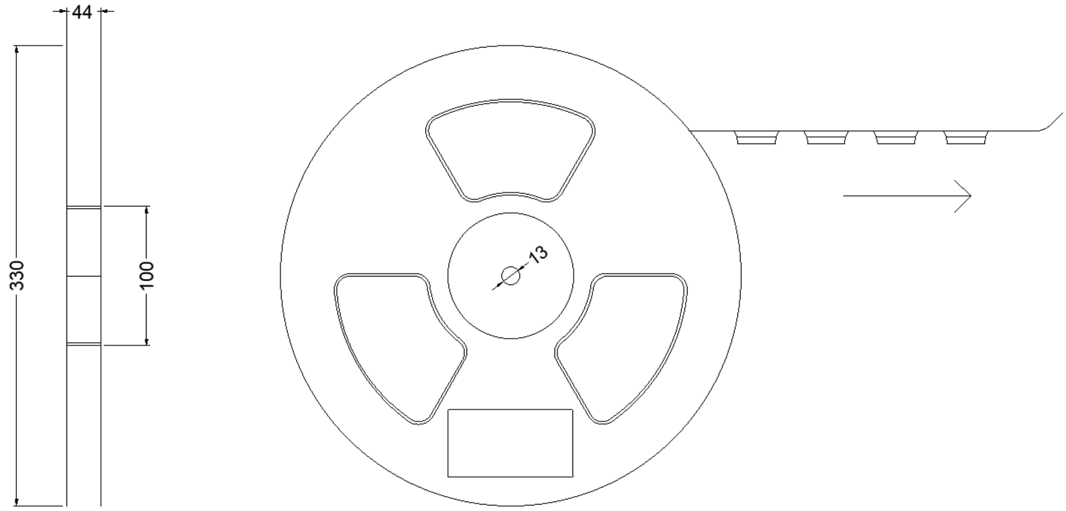


Figure 8: Reel size (unit: mm)8

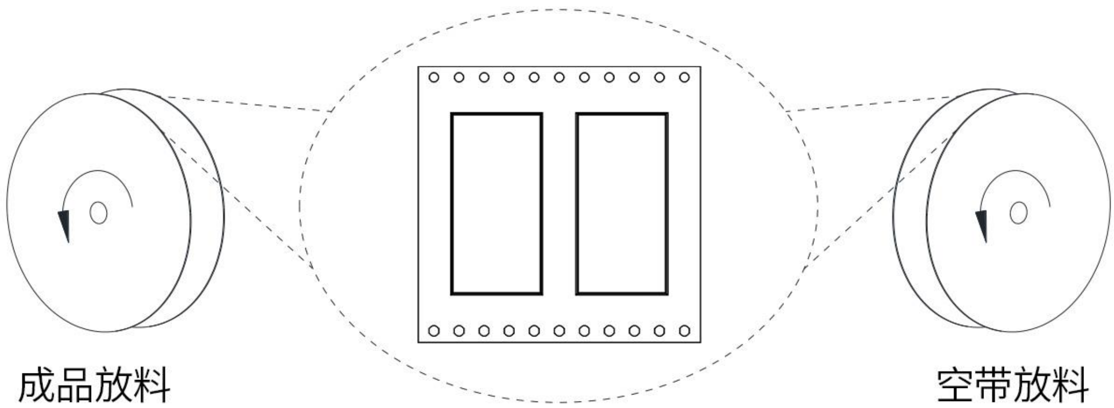


Figure 9: Reel orientation9